



An-Najah National University
Faculty of Engineering and Information Technology

Proposal

AI Customer Support Reply Evaluator

Prepared by :

amr asmar

Nancy Al-Jabbarin

Natural Language Processing

Instructor: Dr. Hamed Abdelhaq

Date: May 5, 2026

Description:

This project proposes an AI system that evaluates customer support replies. The system analyzes responses written by support agents to different customer issues such as delayed orders, damaged products, or refund requests. The model compares each reply with an ideal reference answer and evaluates its quality based on a rubric that includes empathy, accuracy, helpfulness, resolution quality, and professionalism. The system then produces a score (from 0 to 4) along with a short natural language explanation. The goal is to build an automatic and consistent evaluation system using an instruction-tuned language model.

Problem and Motivation

The problem is evaluating customer support replies automatically. We want to check if the reply is good, helpful, polite, and solves the customer's problem.

Rubric-based evaluation is suitable because it gives clear rules for scoring. For example, it checks empathy, accuracy, and professionalism. This makes the evaluation more consistent and less random, and helps the model give both a score and an explanation.

Data Format and Sample Entry

Each dataset entry will be a structured evaluation example designed to assess the quality of a customer support reply. It contains six components: the task (customer issue), a reference (ideal reply), the submission (agent's reply to be evaluated), a rubric (evaluation criteria), a score (0–4), and a rationale (explanation of the decision). This structure ensures consistent and explainable evaluation of customer support responses.

Example:

```
{
  "task": "Customer says the product is damaged and wants a refund.",
  "reference": "We are very sorry that your product arrived damaged. We understand your frustration. We can offer you a full refund or a replacement. Please let us know your preference.",
  "submission": "Return the product if it is damaged.",
  "rubric": {
    "1": "Understands the customer's problem and issue clearly ",
    "2": "Shows empathy and acknowledges customer frustration ",
    "3": "Provides a correct and relevant solution ",
    "4": "Uses polite, clear, professional, and sufficiently detailed response "
  },
  "score": 1,
  "rationale": "The reply is very short, does not show empathy and professional language ."
}
```

Data Source :

The dataset will be **self-authored and manually curated** by our team. We will create customer support scenarios that include different types of customer complaints and inquiries, such as delayed orders, damaged products, wrong-item deliveries, refund requests, billing issues, subscription cancellation, and technical problems.

Public customer support datasets may be reviewed only as inspiration for realistic complaint scenarios and reply styles. However, the final dataset entries will be created and adapted by our team to match the required project format.

For each scenario, we will prepare the required fields: **task, reference, submission, rubric, score, and rationale**. The **task** will describe the customer complaint or support situation. The **reference** will describe the characteristics of an ideal customer support reply. The **submission** will be the agent's reply that needs to be evaluated.

The final dataset entries, rubrics, scores, and rationales will be prepared by our team to ensure that the replies are evaluated consistently based on **empathy, accuracy, helpfulness, resolution quality, and professionalism**.

Scoring Scale :

The scoring scale will range from **0 to 4**. Each score represents how helpful, accurate, empathetic, and professional the customer support reply is.

0 = Unacceptable reply

The reply is rude, misleading, irrelevant, or does not address the customer's issue.

1 = Very weak reply

The reply contains only a very limited useful element, but it mostly fails to acknowledge the issue, show empathy, or provide a solution.

2 = Partially helpful reply

The reply addresses part of the issue, but it is missing important elements such as empathy, clear resolution, accuracy, or a professional next step.

3 = Good reply

The reply is mostly helpful, polite, and relevant, but it still lacks some detail, clarity, or a stronger resolution.

4 = Excellent reply

The reply clearly acknowledges the issue, shows empathy, provides an accurate and helpful solution, and uses a polite and professional tone.